

RYAN CONROY

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Education

Colorado School of Mines

Bachelor of Science in Electrical Engineering – GPA: 4.00

Expected May 2029

Golden, CO

Experience

Mines Formula SAE

Aug 2025 – Present

Electrical Engineer – PCB Design

Golden, CO

- Designed PCBs for both EV and IC Formula SAE cars, taking boards through the entire design cycle from schematic capture and PCB layout to hardware bring-up.
- Collaborated with other subsystems on enclosure design, connector placement, and harness routing to finalize board outlines, mounting, and clearances before PCB layout.
- Verified designs against FSAE electrical rules, manufacturing constraints, and industry best practices to ensure compliance and manufacturability.

Ace Hardware

Apr 2022 – Present

Sales Associate

Anthem, AZ

- Trained new hires on store procedures and POS operations while managing morning stocking/inventory.
- Assisted customers with product questions, delivering effective solutions in a fast-paced retail environment.

Projects

High-Voltage Presence Indicator | *Mines Formula SAE*

- Engineered a hardware-controlled, isolated flyback converter for high-voltage presence detection on a 450V tractive bus, with a precision-tuned threshold circuit targeting 60V turn-on / 50V turn-off per FSAE safety requirements.
- Implemented the isolation boundary to prevent high-voltage faults from propagating to the low-voltage domain.
- Added redundant LED indication (panel-mount and onboard) to ensure the high-voltage status remains visible even if a single indicator fails.

Telemetry Ground Station | *Mines Formula SAE*

- Designed an ESP32-based PCB to receive and transmit lap-timing data over ESP-NOW, SPI, and UART for real-time vehicle telemetry and driver performance analysis.
- Implemented a power distribution system with automatic switching between Raspberry Pi and bench power, buck conversion, fuse protection, and ESD protection.
- Optimized ESP32-S3 placement and PCB ground-plane clearance around the antenna keepout area to preserve RF range for a 200m wireless link.

Current & Voltage Sensor | *Mines Formula SAE*

- Designed a bus-powered PCB measuring up to 150 A of bidirectional current via a Hall-effect sensor, providing vehicle-wide current telemetry.
- Reduced part count and complexity by replacing a switching regulator with an LDO after dissipation analysis confirmed ~160 mW was well within thermal margin, eliminating switching noise near the analog front end.

FPV Quadcopter & Custom Flight Controller | *Personal Project*

- Researched and assembled a 5-inch FPV freestyle drone, independently evaluating and selecting components using manufacturer datasheets.
- Developing a custom flight controller PCB, evaluating MCU, IMU, power, and communication architectures for an embedded flight-control platform.

Technical Skills

Hardware: Schematic capture, PCB layout, DFM, PCBA, board bring-up, DC-DC converters, high-voltage isolation, serial communication, SMD/THT soldering and rework, harness crimping, oscilloscope, multimeter

Tools: Altium Designer, LTspice, Git

Programming: C++, Python

Protocols: UART, SPI, I²C